External Power – General Data

1. General

External 28 Volt DC power is connected to the aircraft through an external power receptacle located on the left side of the rear fuselage, just aft of the cargo door. On aircraft incorporating S.O.O. 6371, the external ground power receptacle has been relocated to the outer metallic structure at approximately station 55.

EXTERNAL power allows ground testing and engine starting procedures to be done without using the aircraft main batteries. EXTERNAL power is available with external power connected to the aircraft, the MASTER switch selected to the ON position, the BUS TIE switch to NORMAL and the POWER SOURCE switch selected to EXTERNAL (Pre Mod 6/2314) or EXTERNAL switch selected to ON (Post Mod 6/2314) (refer to PSM 1-64-2W for DC Generation Electrical Schematics for details). 28 Volt DC from the ground source energizes the external power relay which connects the external power to the aircraft DC buses; the normally open contacts of the main battery relay isolate the battery from the external power (Pre Mod 6/2314). For Post Mod 6/1293 a diode in the external power control circuit protects the aircraft system from an input of reversed polarity.

When either generator is selected on-line, the generator field relay (K5 or K6) is energized disconnecting the external power relay circuit which, in turn, disconnects the external power source from the buses. (During initial start-up, the generator field circuit must be energized, therefore the starting circuit bypasses the appropriate contacts of relay K5 or K6 to maintain external power input).

2. Servicing

A. Application of External Power (Pre Mod 6/2314)

To apply power to the aircraft buses from a ground power source, proceed as follows:

1) Connect 28 Volt DC power source to external power receptacle.

2) Set MASTER switch to ON.

3) Set POWER SOURCE switch to EXTERNAL.

4) Set BUS TIE switch to NORMAL.

5) Check GND PWR voltmeter for 28 Volt DC reading.

B. Removal of External Power (Pre Mod 6/2314)

1) Set MASTER switch to OFF.

2) Set POWER SOURCE switch to OFF.

3) Disconnect ground power source from external power receptacle.

C. Application of External Power (Post Mod 6/2314)

To apply power to the aircraft buses from a ground power source, proceed as follows:

1) Connect 28 Volt DC power source to external power receptacle.

2) Set EXTERNAL switch to ON.

3) Set BUS TIE switch to NORMAL.

4) Check GND PWR voltmeter for 28 Volt DC reading.

D. Removal of External Power (Post Mod 6/2314)

1) Set EXTERNAL switch to OFF.

2) Disconnect ground power source from external power receptacle.